Check the Model

AP Statistics · Unit 2 · Topic 2.11 · B: 2026-11-12 · E: 2026-11-16 · TEACHER KEY

CED TETHER

- ENDURING UNDERSTANDING: VAR-2 The normal distribution can be used to represent some population distributions.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- VAR-2.A Compare a data distribution to the normal distribution model. [Skill 2.D]
- VAR-2.A.1 A parameter is a numerical summary of a population. VAR-2.A.2 Some sets of data may be described as approximately normally distributed. A normal curve is mound-shaped and symmetric. The parameters of a normal distribution are the population mean, μ, and the population standard deviation, σ. VAR-2.A.3 For a normal distribution, approximately 68% of the observations are within 1 standard deviation of the mean, approximately 95% of observations are within 2 standard deviations of the mean, and approximately 99.7% of observations are within 3 standard deviations of the mean. This is called the empirical rule. VAR-2.A.4 Many variables can be modeled by a normal distribution.
- VAR-2.B Determine proportions and percentiles from a normal distribution. [Skill 3.A]
- VAR-2.B.1 A standardized score for a particular data value is calculated as (data value − mean)/(standard deviation), and measures the number of standard deviations a data value falls above or below the mean. VAR-2.B.2 One example of a standardized score is a z-score, calculated as z = (x - μ)/σ. A z-score measures how many standard deviations a data value is from the mean. VAR-2.B.3 Technology, such as a calculator, a standard normal table, or computer-generated output, can be used to find the proportion of data values located on a given interval of a normally distributed random variable. VAR-2.B.4 Given the area of a region under the graph of the normal distribution curve, it is possible to use technology to estimate parameters for some populations.
- VAR-2.C Compare measures of relative position in data sets. [Skill 2.D]
- VAR-2.C.1 Percentiles and z-scores may be used to compare relative positions of points within a data set or between data sets.

Essential question: When does a correct normal calculation justify a conclusion about actual data?

DOK-3 skill: 4.B · **FRQ pattern:** `is-the-normal-model-appropriate-adjudicate-model-vs-data`

DOK rationale: Part (c) separates model arithmetic from a warranted conclusion, tests the model against shape and an observed tail, and explains the consequence of a mismatch.

First take → rules + (a)–(c) → turn in | **DOK:** 1 → 3 | **Minutes:** 45

Teacher does

- Board slide up at the bell. Read Ben's and Ava's lines; do not say whether the model fits.
- At minute 5, read the rules box; have students start (a).
- Board slide back up at minute 33. Circulate; require every model judgment to point to the histogram.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to using or rejecting the model before any discussion.
- Use the printed rules to check both model calculations in (b).

- Finish (a)–(c); complete the exit line; turn in.

Questions to ask

- Which claim is about a normal curve, and which claim is about the 60 actual students?
- How many bars continue to the right after the peak? What would symmetry require on the left?
- How many students does 0.62% of 60 represent? How many do you actually see above 70?
- What must you inspect before typing values into normalcdf?

Adult role

Solo teacher. Circulate 5 pairs. Accept the arithmetic before pressing on the assumption; do not let 'correct z-score' stand in for a model check.

[WARN] LOOK FOR

- Treating the school's assertion of normality as evidence stronger than the histogram.
- Counting the 60–70 bar as above 70.
- Rejecting the model with no named feature or tail comparison.

SCORING PART (c) — E / P / I

Expected elements

- **model-versus-data** (required) — Distinguishes correct model arithmetic from whether it fits these students
- **shape-and-tail** (required) — Uses the long right tail and observed 4/60 about 6.7 percent versus predicted 0.62 percent
- **consequence** (required) — Explains that the model understates long app times and needs checking against the data

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

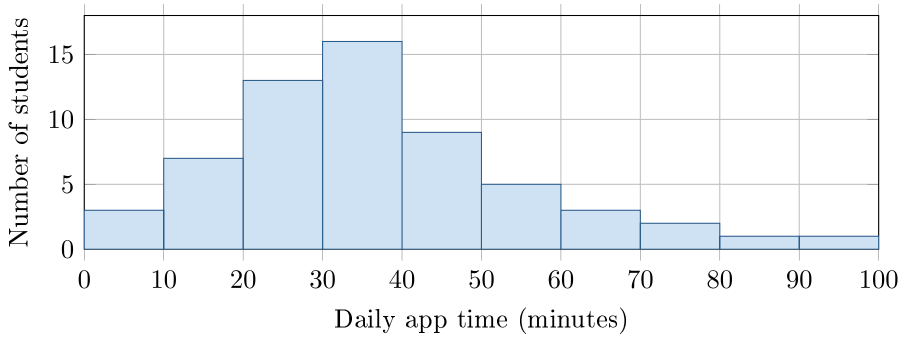
Common mistakes

- Treating a correct z-score as proof that the data are normal
- Saying "not normal" without naming skew, asymmetry, or the long tail
- Counting the 60–70 bin as above 70 or reporting 4 of 60 as 4%

[STAR] TODAY'S PROBLEM — annotated

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes. Table A gives a left area of 0.9938 for $z = 2.5$.

Ben: "About 16% should be above 52 minutes." **Ava:** "For 70 minutes, $z = 2.5$, so less than 1% should be above 70."



FIRST TAKE (5 min)

Do the times look balanced around 40 minutes, or does one side stretch farther? Point to the bars.
Key: Not graded. Expect trust in the stated model; preserve that tension.

Rules callout "USING A NORMAL MODEL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the proposed model's mean μ and standard deviation σ, with units.

Key: $\mu = 40$ min and $\sigma = 12$ min. Approximately normal data should be mound-shaped and roughly symmetric.

DOK 2 **(b)** Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

Key: 52 is 1σ above 40, so Ben's normal-model result is about 16%. For Ava, $P(Z > 2.5) \approx 0.0062 = 0.62\%$. The histogram has $2 + 1 + 1 = 4$ above 70, or $4/60 = 6.7\%$.

DOK 3 **(c)** Would you use Ben's and Ava's model results to describe these students? Use the histogram's shape and the observed versus predicted percent above 70. Explain what a reader could get wrong about long app times.

Key: The arithmetic is correct for the model but not warranted for these data. The histogram peaks at 30–40 and has a long right tail; 4 of 60 (6.7%) exceed 70 versus the model's 0.62%. A calculation inherits its model assumption, so this model underestimates long app times. Check for a mound shape, symmetry, and contradictory skew, gaps, clusters, or outliers before using normal methods.

EXIT

One thing the rules box changed about my first take: ______________________________